

Chi-Sheng (Michael) Chen (陳麒升)

Machine Learning for Physiological Time Series | Clinical Biosignals & Foundation Models | EEG/BCI | Quantum ML

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Nationality: Taiwan | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen builds machine learning for physiological time series along one arc — from representation, to foundation models, to clinical deployment. He develops interpretable, clinician-interrogable frequency-domain and geometric representations of clinical biosignals (FreqLens, SPD Token Transformer) and scales them into biosignal foundation models (Large Cognition Model). His EEG-based depression-treatment model is in routine outpatient use at Taipei Veterans General Hospital's Precision Depression Intervention Center (PreDIC), and at Harvard/BIDMC he builds multimodal EMS trauma-triage pipelines. He has 49 publications, 491 citations (h-index 13, i10-index 16), ICML 2026 Gold Reviewer recognition, and reviewer service across 40 journals (incl. TPAMI, NSR, npj Quantum Information). As a secondary methodological line, he explores hybrid quantum-classical architectures for sequence modeling (QASA, QEEGNet).

RESEARCH & PROFESSIONAL EXPERIENCE

Harvard Medical School & Beth Israel Deaconess Medical Center

MA, USA

Research Affiliate, Advisor: Prof. Dr. Gabriel Brat

Nov 2024 – Present

- Building a real-time EMS trauma-triage pipeline (ASR → LLM-agent event extraction → structured prediction) over **4M+ patient records** using multimodal AI.
- Co-developed webcam-based open-surgical skill assessment (hand keypoint detection; CLINICCAI 2026 poster) and a capture-clock methodology for browser webcam gaze (DEMI 2026 spotlight talk).

WhaleForce (subsidiary of VICI Holdings)

Taipei, Taiwan

Fractional Token Efficiency Research Lead (Advisor)

Aug 2026 – Present

- Advising on LLM token-efficiency research and multi-agent architecture for AI-driven quantitative trading; token-efficiency work cut **LLM token usage by 82%**.

Omnis Labs

Remote

Co-Founder

Sep 2024 – May 2026

- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

Neuro Industry, Inc.

CA, USA

Researcher, Co-Founder & CTO

Mar 2024 – Jan 2025

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 3 research papers: quantum machine learning for EEG signal processing (2) and EEG-to-image diffusion (1).

Department of Computer Science, National Yang Ming Chiao Tung University

Hsinchu, Taiwan

Research Assistant, Advisor: Prof. Chun-Shu Wei

Dec 2023 – Aug 2024

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (32 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc.

Hsinchu, Taiwan

Digital IC R&D Engineer

Dec 2021 – Oct 2023

- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implement.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

Department of Surgery, National Taiwan University Hospital

Taipei, Taiwan

Research Assistant, Advisor: Dr. Shuo-Lun Lai

Jul 2021 – Sep 2021

- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

Department of Psychiatry, Taipei Veterans General Hospital

Taipei, Taiwan

Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li

Sep 2019 – Jun 2021

- Collaborated with researchers and clinicians to identify patterns and trends in brainwave activity related to major depressive disorder.
- Utilized machine learning algorithms to develop predictive models based on real-world brainwave data.

Max Planck Institute for Chemical Physics of Solids (MPI CPfS)

Dresden, Germany

Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao

Jul 2018 – Sep 2018

- Searching new possible unconventional superconductors among Co-based quaternary chalcogenides with diamond like structure $\text{CuInCo}_2\text{A}_4$ / $\text{AgInCo}_2\text{A}_4$ (A = Te, Se, S).

Other: OpenAI (contractor via Mercor) — AI Trainer, RLHF preference data & model-behavior evaluation Mar 2025 – Oct 2025

National Taiwan University (NTU)

Taipei, Taiwan

Master of Science in Biomedical Electronics and Bioinformatics
Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

Sep 2019 – Jun 2021

- Machine learning application on non-stationary time series data.
- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.
- Master's thesis: "EEG Analysis for Prediction of Antidepressant Responses of Transcranial Magnetic Stimulation in Major Depressive Disorder Based on Attentional Convolution Time Series Prototypical Neural Network Model and Classical/Quantum Machine Learning Approaches."

National Chiao Tung University (NCTU)

Hsinchu, Taiwan

Bachelor of Engineering in Electrophysics
Bachelor of Science in Interdisciplinary Science Degree Program

Sep 2015 – Jun 2019

- Researching the oxide processing techniques.
- Researching the biomedical signal processing on traditional Chinese medicine data.

SKILLS

AI/ML: Python, PyTorch, HuggingFace, Scikit-learn, MNE, MLflow, R, MATLAB**Systems:** C/C++, JavaScript, Rust, MySQL, Solidity, Git, Docker, GCP, AWS**Hardware:** Verilog, SystemVerilog, UVM, HSPICE, Genus, Verdi, SG-Lint, Perl**Languages:** Chinese (Native), English (Fluent)

SELECTED PUBLICATIONS (491 CITES, H-INDEX 13, I10-INDEX 16)

First author unless noted. Peer-reviewed unless marked *Under review* or arXiv preprint.

[Physiological Time Series & Clinical Biosignals]

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting" *Under review*, ICLR 2027; arXiv:2602.08768.

Chen, C.S., et al. "A Unified SPD Token Transformer Framework for EEG Classification: Systematic Comparison of Geometric Embeddings" *Under review*, Neurocomputing (Elsevier), 2026; arXiv:2601.21521.

Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model" arXiv preprint (2025), arXiv:2502.17464 — **29 citations**.

Chen, C.S., Wei, C.S. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning" arXiv preprint (2024), arXiv:2406.16910 — **32 citations**.

Chen, C.S., et al. "Exploring the Potential of Electroencephalography Signal-Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis" JMIR Medical Informatics (2025), <https://doi.org/10.2196/72027>.

Chen, C.S., et al. "Exploring the potential of QEEGNet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning" Journal of Signal Processing Systems 98, 5 (2026), <https://doi.org/10.1007/s11265-025-01979-2>.

Chen, C.S., et al. "QEEGNet: Quantum Machine Learning for Enhanced Electroencephalography Encoding" IEEE International Workshop on Signal Processing Systems (SiPS) 2024, doi: 10.1109/SiPS62058.2024.00035, arXiv:2407.19214; poster at QTML 2024, Melbourne (non-archival).

Chen, C.S., Brat, G.A. "Measuring Browser Webcam Gaze Honestly: A Capture-Clock Methodology and Open Reference Implementation" 4th MICCAI Workshop on Data Engineering in Medical Imaging (DEMI 2026), Spotlight talk, arXiv:2608.11566.

Li, C.T., Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation" Journal of Affective Disorders (2023), <https://doi.org/10.1016/j.jad.2023.08.059>. [2nd author; model deployed at Taipei VGH PreDIC]

[Quantum Machine Learning]

Chen, C.S., Kuo, E.J. "Quantum Adaptive Self-Attention for Quantum Transformer Models" Quantum Science and Technology (2026), <https://doi.org/10.1088/2058-9565/ae8b21>.

Chen, C.S., et al. "Unraveling quantum environments: Transformer-assisted learning in Lindblad dynamics" Physical Review A (2025), <https://doi.org/10.1103/gsxk-45mk>.

Chen, C.S., et al. "Quantum Reinforcement Learning-Guided Diffusion Model for Image Synthesis via Hybrid Quantum-Classical Generative Model Architectures" 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP55912.2026.11461991.

Chen, C.S., et al. "Quantum Adaptive Self-Attention for Financial Rebalancing: An Empirical Study on Automated Market Makers in Decentralized Finance" 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP55912.2026.11460972.

Chen, C.S., et al. "Quantum Multimodal Contrastive Learning Framework" 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP49660.2025.10889504, arXiv:2408.13919.

Chen, C.S., et al. "Q-DPTS: Quantum Differentially Private Time Series Forecasting via Variational Quantum Circuits" IEEE GLOBECOM Workshop 2025, arXiv:2508.05036.

Chen, C.S., et al. "Quantum Reinforcement Learning Trading Agent for Sector Rotation in the Taiwan Stock Market" IEEE QCE QCRL Workshop 2025, arXiv:2506.20930.

Chen, C.S., et al. “Improving fine-grained food classification using deep residual learning and selective state space models” PLoS ONE (2025), <https://doi.org/10.1371/journal.pone.0322695> .

Chen, C.S., Liu, H.Y. “Predicting Poets’ Origins from Verse: A Computational Analysis of Regional Linguistic Fingerprints in the Complete Tang Poems” Digital Humanities 2026 (DH2026), Alliance of Digital Humanities Organizations (ADHO), Short Paper.

Lai, S.L., Chen, C.S., et al. “Intraoperative Detection of Surgical Gauze Using Deep Convolutional Neural Network” Ann Biomed Eng 51, 352–362 (2023), <https://doi.org/10.1007/s10439-022-03033-9> . [2nd author]

Lai, S.L., Chou, Y.C., Chen, C.S., et al. “Detection and tracking of a gauze sponge in minimally invasive surgery using a YOLO and R-CNN based model” Medical & Biological Engineering & Computing (2025), <https://doi.org/10.1007/s11517-025-03471-2> . [3rd author]

Liu, H.Y., Chen, C.S. “Gendered Language in Tang Poetry: A Sociolinguistic and Digital Humanities Analysis Based on the Quan Tang Shi.” 16th International Conference of Digital Archives and Digital Humanities (DADH 2025).

Takegami D., et al. “Direct imaging of valence orbitals using hard x-ray photoelectron spectroscopy” Phys. Rev. Res. 4, 033108 (2022), <https://doi.org/10.1103/PhysRevResearch.4.033108> .

Falke J., et al. “Electronic structure of the metallic oxide ReO_3 ” Phys. Rev. B 103, 115125 (2021), <https://doi.org/10.1103/PhysRevB.103.115125> .

Plus 18 further preprints across quantum machine learning, time-series forecasting and medical imaging (full list: Google Scholar).

INVITED TALKS & PRESENTATIONS

Guest Lecture, Wearable Chip Design, NYCU ECE (host: Prof. Sheng-Chieh Huang) — “Quantum Machine Learning for Wearable Biosignals,” Oct 2026.

DEMI 2026 (4th MICCAI Workshop on Data Engineering in Medical Imaging), Strasbourg, France — spotlight talk, Oct 2026.

CLINICCAI 2026 (Clinical Day at MICCAI 2026), Strasbourg, France — poster, presenting author, Sep 2026.

Guest Lecture, Maker Club, NYCU, Hsinchu — “AI, Robotics and Makers in Agent Era,” Sep 2026.

Guest Lecture, Advanced Compilers, NTU CSIE (host: Prof. Shih-Wei Liao) — “Introduction to Classical/Quantum Biosignal Processing,” Sep 2026.

Digital Humanities 2026 (DH2026), Daejeon, South Korea — two short-paper presentations, Jul 2026.

IEEE ICASSP 2026, Barcelona, Spain — two oral presentations, May 2026.

NVIDIA GTC 2026, San Jose, CA — two poster presentations, Mar 2026 (**Top 8 Finalist**).

Invited Seminar, Dept. of Applied Physics, Tunghai University, Taichung — Dec 2025.

IEEE Quantum Week (QCE 2025), Albuquerque, NM — paper presentation, Aug 2025.

IEEE ICASSP 2025, Hyderabad, India — oral presentation, Apr 2025.

QTML 2024 (8th Int. Conf. on Quantum Techniques in Machine Learning), Melbourne, Australia — poster, QEEGNet, Nov 2024.

PROFESSIONAL SERVICE

Program Committee: QNRL Workshop @ IEEE WCCI 2026

Conference Reviewer (2026–2027): **NeurIPS**, **ICML** (Gold Reviewer — top tier), **AAAI**, **KDD**, **MICCAI**, **ICASSP**; NLDL 2027; IEEE QAI 2026

IOP Trusted Reviewer (2026), recognized by the Institute of Physics Publishing for quality and commitment in peer review.

Journal Reviewer (2025–present) — **40 journals**, incl. IEEE TPAMI, Oxford National Science Review, IEEE TNNLS, Elsevier Neural Networks, IEEE/ACM TASLP, IEEE JBHI, IEEE TNSRE, IEEE TQE, npj Quantum Information, Scientific Reports, ACM TORS (full list: [chishengchen.github.io](https://github.com/chishengchen)).

AWARDS & ACHIEVEMENTS

2026 NVIDIA GTC Top 8 Finalist: “GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB.”

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

2023 Certificate of the 20th National Innovation Award (Taiwan), clinical research category — for the EEG treatment-response model now deployed at Taipei VGH PreDIC.

2019 International Blockchain Olympiad (IBCOL): World Finalist competition with 100+ teams.

2017 Microsoft International Imagine Cup : 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.